Problem

1. Water Jug Problem – Problem Statement

You are given two water jugs: one with a **4-gallon capacity** and another with a **3-gallon capacity**. Neither jug has any measurement markings. A water pump is available to fill either jug completely. Water can also be poured from one jug to another or emptied onto the ground. The objective is to obtain **exactly 2 gallons of water in the 4-gallon jug** using the allowed operations.

---

2. Mars Rover Intelligent Agent – Problem Statement

A **Mars Rover** is an autonomous intelligent agent designed to explore the surface of Mars. It collects scientific data by navigating different terrains, capturing images, analyzing rocks and soil samples, and transmitting information back to Earth. Analyze the Mars Rover by identifying its percepts, operating environment, possible actions, performance measures, and the most suitable agent architecture.

---

3. 8-Queens Problem – Problem Statement

The **8-Queens problem** is a classic Artificial Intelligence search problem. The objective is to place **eight queens on an 8 × 8 chessboard** so that no two queens attack each other. A queen can attack any other queen placed in the same **row, column, or diagonal**. Formulate the problem by identifying the initial state, state space, operators, goal state, and suitable search strategy.

---

4. OLA Cab Booking – Problem Statement

A customer wants to travel from one location to another using the **OLA Cab Booking** mobile application. The customer enters the pickup and destination locations and selects a preferred cab type such as **Mini, Micro, Sedan, Prime, or Shared** based on comfort and budget. The system must identify an appropriate problem-solving agent to process the request and generate a sequence of actions that successfully books the cab and completes the journey.

---

5. Uniform Cost Search (UCS) – Problem Statement

A logistics company operates a network of warehouses connected by routes with different travel costs. Each route has an associated cost based on fuel consumption and travel time. The objective is to determine the **least-cost path** from the **starting warehouse (S)** to the **destination warehouse (G)** using the **Uniform Cost Search (UCS)** algorithm on the given weighted graph. The algorithm should always expand the node with the lowest cumulative cost until the destination is reached.